

Module 11: Generic Lists

Objective: Creation and handling of generic lists

System requirements: .Net 5.0

Tools: Visual Studio 2019

Duration: ~30min

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1 Generic vehicle lists

The goal of this lab is to fill lists with vehicles from the previous exercises in the Main() method and run through them automatically.

1. First create a Queue<Vehicle> and a Stack<Vehicle> and fill them with 10 vehicles each. The selection of the vehicle type should be random.
2. now try to load the foremost element of the queue with the topmost object of the stack.
 - a. If this is successful both elements are to be inserted into a Dictionary<Vehicle, Vehicle> as a new entry.
 - b. If unsuccessful, both elements shall be discarded.
3. Repeat this until stack or queue are empty.
4. output the dictionary in the form "vehicle A has loaded vehicle B".

```
Loading of 'Boeing_S9' onto 'Titanic_Q0' successful.
Loading of 'Titanic_S7' onto 'Titanic_Q2' successful.
Loading of 'VW_S4' onto 'Titanic_Q5' successful.
Loading of 'Boeing_S2' onto 'Titanic_Q7' successful.
-----Cargo list-----
'Titanic_Q0' has loaded 'Boeing_S9'
'Titanic_Q2' has loaded 'Titanic_S7'
'Titanic_Q5' has loaded 'VW_S4'
'Titanic_Q7' has loaded 'Boeing_S2'
```